Amazon EC2 Exercises

Exercise 1

Launch and Connect to a Linux EC2 Instance

In this exercise you will launch a new Linux instance, log in with SSH and install any updates available for Linux Ubuntu v16.04 AMI.

1- Launch an instance in the Amazon EC2 Management Console.
2- Chose the Amazon Linux AMI.

3- Choose the t2.micro instance type.

4- Launch the instance in either the default-VPC or EC2-Classic.

5- Assign the instance a public IPv4 address.

6- Add a tag to the instance of Key: Name, Value: Exercise 1

7- Create a new Security Group called 'mySG'

8- Add a rule to 'mySG' allowing SSH access from IP address of your workstation/local-laptop (www.WhatsMyIP.org) is a good way to determine your IP address.

9- Launch the instance.

10- When prompted for a Key-Pair, choose the Key-Pair you already have or create a new Key-Pair and download the Private-Key.

Amazon generates a keyname.pem file, and you will need a terminal-emulator tool (SSH tool) to log-into your EC2 instance. Use Putty/XShell/MobaXtern to access your EC2 machine from your local-laptop.

11- SSH into the instance using the Public IP address, the user-name will be 'ubuntu' (for Amazon Ubuntu AMIs and the keyname.pem file.

12- From the command-line prompt, run command: sudo apt-get update

13- Close the SSH window and terminate the instance.

Exercise 2

Confirm that the Instance Stores are LOST when an Amazon EC2 Instance is Stopped :-

In this exercise you will observe that the data on an Amazon EC2 instance store is lost when the instance is stopped.

1- Launch an EC2 instance in the AWS Management Console.
2- Choose Ubuntu v16.4 AMI from the AWS Market Place.
3- Choose t2.micro instance type.
4- Launch the instance in either default-VPC or EC-Classic.
5- Assign the instance a Public IPv4 address.
6- Add a tag to the instance, Key: Name, Value: Exercise 2
7- Use the Security Group from 'Exercise 1'
8- Launch the EC2 instance
9- Use Key-Pair from 'Exercise 1'
10- SSH into the instance using the Public IP address, the user-name will be 'ubuntu' (for Amazon Ubuntu AMIs and the keyname.pem file.
11- Create a new directory at the root location using command: sudo mkdir myTestDirectory
> ls (ls command will list all directories at the current location on our Linux machine.)

12- Close the SSH window
13- In the Management Console set the state of the EC2 instance to 'Stopped.'
14- Once the instance has stopped, start it again.
15- Log back into the EC2 instance using SSH.
16- Now again run the 'ls' Linux-Command to list all directories at the root-location. You will notice that our 'myTestDirectory' directory is gone.
17- End the SSH session and terminate the instance.

Exercise 3

Launch an Amazon EC2 Spot Instance

1- In the Amazon EC2 Management Console, go to Spot Request page.
2- Look at the pricing history for m3.medium, especially the recent price.
3- Make a note of the most recent price and the Availability Zone.
4- Launch an instance in the Amazon EC2 Management Console.
5- Choose the Amazon Linux AMI (Ubuntu)
6- Choose the t2.medium instance type.
7- On the Configure Instance page, request a Spot Instance.
8- Launch the instance in either the Default-VPC or EC2-Classic (Note: Default-VPC will define the Availability Zone for the instance.)

9- Assign the instance a Public IP Address.
10- Request a Spot Instance and either a bid few cents above the recorded Spot Price.
11- Finish launching the instance.
12- Go back to the Spot Request page.

Watch your request. If our bid was high enough you should see it change to Active and an instance ID appears.

13- Find the instance on the instances page of the Amazon EC2 Management Console.
14- Once the instance is running, terminate it.

Exercise 4

Access Instance Metadata.

1- Launch an instance in the AWS Management Console.
2- Choose the Amazon Linux AMI (Ubuntu.)
3- Choose t2.micro instance type.
4- Launch the instance in either in Default-VPC or EC2-Classic.
5- Assign the instance a Public IP address.
6- Add a tag to the instance of Key: Name, Value: **Exercise 4**
7- Use the 'mySG' Security Group.
8- Launch the instance.
9- Use Key-Pair from 'Exercise 1.'
10- Connect to the instance via SSH and using Public IP address, the user-name: ubuntu, and the Key-Pair: Keyame.pem file.
11- At the Linux command prompt, retrieve a list of the available metadata by typing: http://169.254.169.254/latest/meta-data/

12- To see a value, add the name to the end of the URL. For example, to see the 'Security Groups', type: http://169.254.169.254/latest/meta-data/security-groups

13- Try other values as well. Names that end with a '/' indicate a longer list of sub-values.
14- Close the SSH window and terminate the instance.